

technologies that will better our planet. Of course, Digit could probably publish a Fast Track on every one of the many technologies that are mentioned here, and perhaps we will. However, due to space constraints, we're only going to whet your appetite by covering what we think you should know about the invisible technologies that surround us, in no particular order of importance.

Networks

OK! So we lied. In the sections that follow, there may not be any particular order, but networks come first, because without them, everything's just a dead, useless chunk of silicon and metal.

When you think about it, absolutely any technology that you're exposed to is network driven. Regardless of landline or mobile, if you call someone, it's because of a network. If you browse the web, that's a network. Your Bluetooth headset that connects with your phone, booking a ticket, paying your bills, withdrawing money from a bank/ATM, email, SMS, the electricity that powers your fan...all network-driven.

Networks aren't new. They're as old as the Earth, and older than mankind. The very first multi-celled organism that came out of the primordial soup on Earth almost four billion years ago made use of networking. Cells communicate chemically with one another, and yes, all of us are examples of the most complex networks in nature. In fact, the network that is you – trillions of cells that talk to one another to form organs, multiple organs connected by a nervous system, all hooked in to your CPU (brain) – is more complex than the network of every single PC, gadget or device on Earth. The Internet? Bah humbug, that's about as complex a network as your big toe.

As is with a lot of technologies we take for granted today, or even those we

marvel at, all we're really doing is making rather shabby replicas of nature. The next time you marvel at your DSLR or your spanking new 4K TV (that you're still paying for), remember, your eye has a resolution of about 500 to 600 megapixels (that's 600,000,000 pixels!), and the eye-brain network is more complex than all of the chips in all of the TVs ever made, collectively!

Now that we're done praising good ol' networked you, let's get down to the various invisible networks that really power our world.

Financial Networks

Sure, this may be just a bunch of computers and servers connected to one another, just like any other network, but it's one of the most important. Money makes the world go round (our apologies to gravity), and without networks, we'd have no way of knowing how much we have. Actually, it's already hard enough to know how much you have. Take for example estimates that put the amount of cold hard cash in the world to be about four trillion USD, and now compare that to the value of all the countries on earth – about 70 trillion USD. You may only have ₹100 in your pocket, and another ₹1 lakh in your bank, but you are probably worth much more. Our favourite way to look at it is to imagine that although this issue cost you ₹200, the value you get could be orders of magnitude more! Value is always more than the amount of cash in the world, which is why if every person in India decided to withdraw all of their money from banks at the same time, there'd be riots – because most would be turned away empty handed.

So how does it work? You earn a salary, it's deposited into an account, you withdraw some amount of money, the bank has details of how much you're allowed to take out, and deducts the correct amount



The people behind the scenes

from your account. All of this involves data flying across a complex network with about the same level of security as used in top secret government facilities, and often uses the networks of multiple banks simultaneously – when you use an ATM of another bank for example. Now multiply that by a few billion, and you can understand the complexity of the world's financial systems.

Cheques, DDs, online transfers, currency exchange rates, etc. all use a similar method of keeping tabs on money, and the value it has, and are thus, almost entirely dependent on networking.

Neural networks, Google and a bank



Ever wondered how your brain functioned? Well, so did many of the world's top scientists. A handful of them got together and decided to start the Human Brain Project. A fascinating initiative that aims to recreate the entire human brain by mapping the neural network to a computer simulation. Read about all this and more at <http://dgit.in/CSJuly1>



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